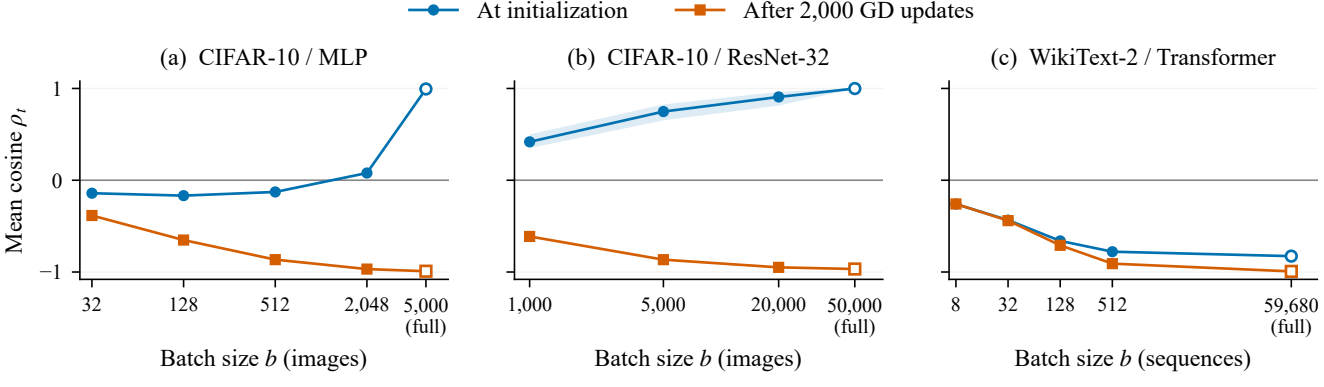

At initialization
After 2,000 GD updates
(a) CIFAR-10 / MLP
(b) CIFAR-10 / ResNet-32
(c) WikiText-2 / Transformer
Mean cosine ρ_t
1
0
−1
32
128
512
2,048
5,000
(full)
1,000
5,000
20,000
50,000
(full)
8
32
128
512
59,680
(full)
Batch size b (images)
Batch size b (images)
Batch size b (sequences)